

Example 3-7: Simplify the Boolean function:

$$F(A, B, C, D, E) = \Sigma(0, 2, 4, 6, 9, 11, 13, 15, 17, 21, 25, 27, 29, 31)$$

AB \ CDE	000	001	011	010	110	111	101	100
00	1			1	1			1
01		1	1			1	1	
11		1	1			1	1	
10		1					1	

$$F = BE + AD'E + A'B'E'$$

Note: Do upto 4 variable K-Map for now

3.5 Product of Sums Simplification

- The procedure for obtaining a minimized function in product of sums follows from the basic properties of Boolean functions. The 1's placed in the squares of the map represent the minterms of the function.
- The minterms not included in the function denote the complement of the function. From this we see that the complement of a function is represented in the map by the squares not marked by 1's.
- If we mark the empty squares by 0's and combine them into valid adjacent squares, we obtain a simplified expression of the complement of the function, i.e. of F' . The complement of F' gives us back the function F .
- Because of the generalized De Morgan's theorem, the function so obtained is automatically in the product of sums form.

Example 3-8: Simplify the following Boolean function in (a) sum of products

and (b) product of sums.

$$F(A, B, C, D) = \Sigma(0, 1, 2, 5, 8, 9, 10)$$

AB \ CD	00	01	11	10
00	1	1		1
01		1		
11				
10	1	1		1

The sum of product form is:

$$F = BD + BC + AC'D$$